prd.md

> **CivicSync · Product Requirements Document** Version 1.0-rc1 · 28 Oct 2026 · Status: Round 1 of the `loop.md` review done; awaiting team sign-off Owner: UX & Ops Lead (Somil Tomar) · Co-owner: Team Leader (Aayush) Read with: `trd.md`, `rules.md`, `design.md`, `memory.md` (assumptions are numbered **A-xx**, decisions **D-xx**, open questions **Q-xx**, source conflicts **C-xx**).

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1. Why we are building this

When a road hazard shows up on someone's daily route, there is no easy way to tell the city, and no way to know whether anyone heard. Our pilot survey and the people who answered it say the same few things again and again: *"I didn't know where to report," "it's too complicated,"* and *"I didn't think anyone would respond."* On the city's side, complaints arrive as phone calls, paper notes and portal forms with typed addresses, no photo, no exact spot, and a lot of repeats. Nobody can tell which pothole is the dangerous one.

CivicSync is a phone app plus a municipal dashboard. A citizen points the camera at a hazard and sends it; the app fills in the place, the time and its best guess of the hazard type. The city sees one ranked list of real, de-duplicated problems and moves each one through a clear status that the citizen can follow.

1.1 What the evidence says

We treat the survey as *qualitative evidence*. We do not compute new totals from it.

- **Numbers quoted in our Screen-1 deck** (not recomputed here; see **C-01** in `memory.md`): 104 respondents (Agra 47, Jodhpur 48, other cities 9); 71% met a hazard in the last 30 days; 82% of those did not report; 53% would report if it took under 30 seconds. Reasons for not reporting: didn't know where (59%), expected no reply (46%), too complicated (41%).
- **National context** (MoRTH, as cited in the proposal): pothole-related deaths rose from 1,555 in 2020 to 2,385 in 2024.
- **Themes in the attached response export (92 rows):** potholes and water-filled potholes in monsoon; poor-quality repairs; missing warning signs on highways; poor visibility at night; stray cattle; wrong-side driving; congestion; a repeated wish to see an update after reporting. Respondents' own words included "I have but I didn't get any update if it's resolved or not."
- **Caution:** some rows in the export look low-effort or bulk-filled (for example, a block of ten answers inside about twelve seconds). We use them for themes only until the sheet is cleaned (**C-01**).

1.2 The three barriers we design against

Barrier people told us about	What the product does about it
"I don't know where to report."	One app, one button, opens straight to the camera. No ward, form or department to choose.
"It's too complicated."	Photo → confirm → send. Place, time and hazard type are filled in for the person. Target: about 30 seconds or less from opening the app (see §6).
"No one will respond."	Every report gets a visible status: Reported → Verified → Assigned → Resolved. Officers are held to response times (`rules.md`).

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2. Who it is for

These personas are composites built from the survey themes and the team's field visits. They are not real individuals.

Persona	Who they are	What they need	What worries them
Imran, the two-wheeler commuter (Jodhpur)	Rides daily on a scooter. Two-wheeler riders were the largest group in our survey.	A very quick way to flag a hazard before someone gets hurt.	Wasting time on a form; hearing nothing back.
Meena, the pedestrian and bus passenger (Agra)	Walks and uses public transport.	To see whether the problem near her lane is already known.	Not knowing whether anyone cares.
Kavita, the Ward Officer	Gets complaints for one or more wards.	A ranked list she can trust, with photo and exact spot, and no repeats.	Being blamed for pothole backlogs she can't see clearly.
Deepak, the Crew Lead	Leads the repair crew.	Clear jobs: where, what, how bad, and a place to attach the "fixed" photo.	Vague addresses; being sent to the same spot twice.
Mr. Sharma, the Zonal Engineer / City Admin	Oversees several wards.	Heatmaps, overdue lists, and numbers he can show his seniors.	Having no evidence for budget requests.

The six system roles that map to these people are defined in `security.md` §4: **Citizen, Ward Officer, Crew Lead, Zonal Engineer, City Admin, Platform Admin**.

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3. Goals and non-goals

3.1 Goals (for the 75-day prototype and pilot readiness)

1. **G1 – Make reporting easy.** A first-time user can send a good report in about 30 seconds or less.
2. **G2 – Make reports useful.** Every report arrives with a photo, an exact location, a time, a hazard class and a starting severity.
3. **G3 – Cut the noise.** Repeat reports about the same hazard are grouped into one ticket automatically.
4. **G4 – Show people what happens next.** Citizens can see their report's status; officers have response-time targets.
5. **G5 – Stay cheap.** Pilot cloud cost stays under **$50/month** (`tooling.md` §6).
6. **G6 – Reach TRL 5–6 by Day 75** with measured results from a closed beta in Agra and Jodhpur.

3.2 Non-goals

- Replacing a city's repair contracting, budgeting or payment systems.
- Detecting hazards automatically from moving vehicles (dashcam mode).
- Becoming a general civic-complaints app (see §8).

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4. Scope of version 1 (pilot build)

4.1 Citizen app (Android first, iOS if a test device is available; A-09)

ID	Requirement	Priority
FR-C1	Open straight to the camera; capture one photo.	Must
FR-C2	Capture GPS location and time automatically; show a small map pin the person can nudge.	Must
FR-C3	Suggest a hazard class on the phone, with no internet needed: Pothole, Open drain or missing cover, Broken or cracked road, Waterlogging, Other.	Must
FR-C4	Let the person correct the class and optionally pick a size (Small, Medium, Large, Not sure).	Must
FR-C5	Save the report offline and send it when a connection returns.	Must
FR-C6	Show "My reports" with the four statuses and plain-language updates.	Must
FR-C7	Show "Already reported nearby" before sending, so people can add their voice instead of filing a copy.	Should
FR-C8	Hindi and English UI.	Should (Hindi strings by Sprint 4)
FR-C9	Push notifications when a status changes.	Should
FR-C10	Blur faces and number plates on the phone before upload, as best-effort.	Should
FR-C11	"Still there?" check on fixed tickets, so citizens can confirm or reopen.	Could

4.2 Municipal dashboard (web)

ID	Requirement	Priority
FR-M1	Secure sign-in with roles; each person sees only their wards.	Must
FR-M2	Ranked priority queue (Critical, High, Medium, Low) with filters.	Must
FR-M3	Map view with ticket pins and a density view of where hazards cluster.	Must
FR-M4	Ticket detail: photos, class, severity, number of reports, timeline.	Must
FR-M5	Status actions: Verify, Assign, Resolve (with an "after" photo), Reject (with a reason).	Must
FR-M6	Duplicate review: confirm or split suggested merges.	Should
FR-M7	Overdue and escalation list.	Should
FR-M8	Simple analytics: tickets by class, ward, time to fix. CSV export for City Admin.	Could

4.3 Back end and AI (no user interface)

Ticket API, ward lookup, duplicate grouping, priority scoring, notification triggers and the on-device model. Details in `trd.md` and `architecture.md`.

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5. User stories and acceptance criteria

Format: *Given / When / Then*. Targets in brackets are set in `trd.md` and `rules.md`.

US-C1 – Report in a few taps (FR-C1–C4) As Imran, I want to photograph a pothole and send it quickly, so that I don't lose my ride time.

- *Given* the app is installed and permissions are granted, *when* I open it, *then* the camera is ready without any menus.
- *Given* I have taken a photo, *when* the screen loads, *then* the place, the time and a suggested hazard type are already filled in.
- *Given* I tap Send, *then* I see a clear "Thanks, we've got it" within 3 seconds on 4G [TRD-P3].
- A tester doing this for the first time reaches "Thanks" in 30 seconds or less on at least 80% of attempts [PRD-M1].

US-C2 – Report with no signal (FR-C5) As Imran riding through a weak-signal stretch, I want the report saved so I don't have to redo it.

- *Given* no connection, *when* I tap Send, *then* the app says "Saved. We'll send this when you're back online," and keeps the photo and place.
- *Given* the connection returns, *then* the report is sent without me opening the app and my report list updates.
- Saved reports are kept for up to 7 days [TRD-O2].

US-C3 – Correct the app's guess (FR-C4) As Meena, I want to change the hazard type if the app gets it wrong.

- *Given* a suggested class, *when* I tap it, *then* I can pick another in one tap.
- The final class stored is the person's choice; the model's guess is stored separately for learning (`rules.md` §8).

US-C4 – Know what happened (FR-C6, FR-C9) As Meena, I want to see where my report stands.

- *Given* I've sent a report, *then* "My reports" shows one of: Received, Checked, Crew assigned, Fixed (the four statuses in plain words, `design.md` §6).
- *Given* an officer changes the status, *then* I get a push notification within 1 minute (online devices).
- Rejected reports show a kind reason, never just "Rejected".

US-C5 – Don't make duplicates (FR-C7) As Meena, I want to know if my lane's pothole is already reported.

- *Given* an open ticket within 25 m with a compatible class, *when* I finish a photo, *then* the app asks "This looks already reported. Add your photo to it?" and the report is attached to that ticket either way.

US-C6 – Protect my privacy (FR-C10) As any citizen, I want people's faces and car plates not to be shared.

- *Given* a photo contains a detected face or plate-like text, *then* that area is blurred on the phone before upload (best-effort, see `security.md` §5).

US-M1 – See what matters first (FR-M2, FR-M3) As Kavita, I want the most dangerous hazards at the top.

- *Given* I sign in, *then* I see tickets from my wards only, sorted by Priority Score, highest first [`rules.md` §1].
- *Given* a ticket has been reported by several people, *then* it shows the count and ranks higher.

US-M2 – Act and update (FR-M5) As Kavita, I want to verify, assign and resolve in a few clicks.

- *Given* a Reported ticket, *when* I click Verify, *then* it becomes Verified and the citizen is notified.
- *Given* a ticket is Assigned, *when* Deepak closes it, *then* he must attach an "after" photo before Resolved is allowed.

US-M3 – Merge repeats safely (FR-M6) As Kavita, I want suspected repeats grouped, but I want to be able to undo a wrong merge.

- *Given* two reports 25–50 m apart, *then* they appear in "Possible duplicates" and I can merge or keep apart.

US-M4 – Stay on top of deadlines (FR-M7) As Mr. Sharma, I want to see what is overdue.

- *Given* a ticket passes its stage deadline [`rules.md` §4], *then* it appears in the Overdue list and the right officer is alerted.

US-A1 – Safe staff access (FR-M1) As Platform Admin, I want only approved staff to sign in.

- Staff accounts are invite-only, use two-step verification, and can be disabled in one action (`security.md` §4).

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6. Success metrics

All targets below are **assumed pilot targets** (**A-05**) for the team and judges to challenge. They are not claims about current performance.

ID	Metric	Target	How we measure	When
PRD-M1	First-time reporters who finish within 30 s	⩾ 80%	Timed beta sessions + app timing events	Sprint 5
PRD-M2	Capture-to-"Thanks" on 4G	< 3 s	Instrumented build on a reference phone	M2, M5
PRD-M3	On-device inference time (p95)	< 150 ms	Benchmark on reference device (**A-04**)	M3
PRD-M4	Classifier quality on the Agra/Jodhpur held-out set	Top-1 ⩾ 80%, macro-F1 ⩾ 0.75	Confusion matrix on the field test set	M3, M5
PRD-M5	Duplicate grouping precision (officer review sample)	⩾ 90% correct merges	Officer "split" rate in beta	Sprint 5
PRD-M6	Beta reach	⩾ 40 distinct testers; ⩾ 200 reports across Agra and Jodhpur	Dashboard counts	Sprint 5
PRD-M7	Officers' median time-to-verify during beta	Within the SLA in `rules.md` §4	Ticket timeline	Sprint 5
PRD-M8	Pilot cloud cost	< $50 per month	Billing and budget alerts	Continuous
PRD-M9	Citizens who say status updates were "clear" (beta survey)	⩾ 70%	Short post-beta survey	Sprint 5

If PRD-M4 is not met, we fall back to the three-class set in `rules.md` §8.4 and re-measure (the proposal's fallback).

7. Release plan (maps to sprints)

Sprint	Days	What citizens or officers get
1	1–15	Clickable prototype; app that captures a photo with place and time (M1)
2	16–30	Reports reach the cloud; offline save and send (M2)
3	31–45	Hazard suggestion and starting severity on the phone (M3)
4	46–60	Municipal dashboard, statuses, duplicate grouping (M4)
5	61–75	Closed beta, measurements, final demo (M5, TRL 5–6)

The full calendar, including DTU-SEVA gates, is in `phases.md` and `roadmap.md`.

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8. Out of scope for version 1

People told us about many road problems. We will not pass these on yet. The app will say so kindly (`design.md` §7).

- **Traffic behaviour and congestion:** wrong-side driving, jams, rickshaw bunching, level-crossing waits.
- **Stray animals** on roads.
- **Garbage and litter** (this belongs to sanitation channels such as Swachhata).
- **Emergencies:** the app is *not* an ambulance or police channel and says so.
- Payments, contractor bidding, or repair budgeting.
- Public map with photos. (Citizens see their own reports and nearby open hazards without photos.)
- Automatic estimate of damage size from the photo (the size chip is manual in v1; see **D-09**).
- Phone-number login (SMS costs and registration, **D-06**).
- Missing or unsafe road signs and badly planned speed breakers: accepted only under **Other** with a note, not auto-detected.

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9. Risks that affect the product

Risk	Effect	Plan
Not enough labelled images (especially drains and waterlogging)	Weak classifier	Combine public sets with field photos; fall back to three classes
Officers don't use the dashboard	No visible response, and citizens lose trust	Involve a ward officer from Sprint 4; keep actions to one click
Free-tier limits hit (storage, pause)	Outages in demo	Pro plan before beta; image compression; retention (`rules.md` §7)
Misuse or fake reports	Wasted crew visits	Rate limits, photo checks, officer verification (`security.md` §3)
Schedule overlap with DTU–SEVA stage dates	Judges see an unfinished build	Demo-slice plan in `phases.md` §3 (**C-10**)

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10. Document acceptance criteria

This document passes review when it meets the checklist for `prd.md` in `loop.md` §3.